INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

FELT, SHEET, POLYESTER, NEEDLE PUNCHED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a needle punched polyester felt sheet material.

1.2 Classification. The felt sheet is available in one (1) type of polyester needle punched. (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

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AMSC N/A FSC 8305

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM20A - Test Method for Fiber Analysis: Quantitative

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1777 - Standard Test Method for Thickness of Textile Materials
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3511/D3511M - Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Brush Pilling Tester
ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5103 - Standard Test Method for Length and Length Distribution of Manufactured Staple Fibers (Single-Fiber Test)

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Material.

3.3.1 Fiber blend. The felt shall be manufactured by using a blend of 75 percent, 6 denier, 3 inches staple length, natural color polyester and 25 percent, 3 denier, 2-inch staple length, black color polyester. Testing shall be as specified in 4.5. Reprocessed polyester shall not be used.

3.3.2 Needle-punched felt. The felt shall be made by using the needle-punched process by vertically needling fibers to desired thickness, density and performance requirements.

3.4 Color. The color of the finished felt shall be grey, resulting from the fiber blend specified in 3.3.1, when examined on the urethane coated finished side.

3.5 Finish. The felt shall have an evenly applied urethane coating to one side adding 3.0 to 4.0 ounces per square yard to the finished felt.

3.6 Physical requirements. The finished felt shall conform to the physical requirements listed below in Table I, when tested as specified in 4.5.

TABLE I. Physical requirements.

Characteristic	Requirement
Weight, oz./yd^2 (after coating)	
Minimum	33.0
Maximum	40.0
Thickness, (inches)	
Minimum	0.265
Maximum	0.335
Breaking strength, pounds, (min.)	
Machine direction	390
Cross machine direction	580
Splitting resistance, pounds, (min.)	
Machine direction	29
Cross machine direction	33
Fuzz resistance 1/	4- slight fuzzing

1/ Fuzzing is considered to be a hairy effect protruding the fabric caused by broken fibers or filaments.

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3.6 Width. The felt shall be trimmed and the minimum width after trimming shall be as specified (see 6.2).

3.7 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the felt shall be furnished in continuous lengths, in roll of not less than 25 yards and not greater than 55 yards. Each roll shall contain no more than two (2) pieces and no piece shall be less than three (3) yards in length.

3.8 Fiber identification. Each roll of the finished felt shall be labeled or ticketed for fiber content in accordance with the 16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act.

3.9 Identification tickets. For Government procurement only, unless otherwise specified (see 6.2) each roll of felt shall have a firmly attached ticket to the edge at the inner edge of the felt roll. The ticket shall be legibly marked with water insoluble ink and shall include the following information:

Nomenclature, thickness
Color
Federal stock number
Supplier
Contract number and date
Specification number and date
Name of contracting office
Supplier's piece number
Gross yards
Net yards

3.10 Workmanship. The finished felt shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5 and 4.6.

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4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. Each roll of felt in the sample shall be examined yard by yard on the polyurethane coated side for defects listed in Table II. All the defects found shall be counted regardless of their proximity to each other, except where two more defects represent a single local condition of the felt, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one (1) for every lengthwise yard or fraction thereof in which it occurs. Defects are scored when clearly visible at normal inspection distance of three (3) feet.

TABLE II. List of defects.

Any hole, cut, or tear.
Crease or wrinkle, hard embedded.
Any spot, stain, or streak.
Thin spots, or thin edges.
Broken needles embedded in felt.
Urethane coating omitted from any area on the felt.
Color not as specified.
Width not as specified.
Objectionable odor.
Label missing, incorrect or illegible.
Required information missing from the label.
Not packaged in accordance with the contract or purchase order.

4.4.1 Roll and fiber identification and marking examination. During the yard-by-yard examination each roll of felt in the sample shall be examined for fiber and roll identification and scored as a defect if missing or not legible.

4.4.2 Length examination (individual and total yardage). During the yard-by-yard examination, each roll of felt in the sample shall be examined for length. Any length found to be less than the minimum or more than the maximum specified or more than two (2) yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two (2) or more rolls in the sample are defective with respect to length if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets.

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4.5 Material testing. The felt shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 1/2-yard full width for felt 40-inches wide or more and one (1) yard full width for felt under 40-inches wide. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Material tests.

Characteristic	Requirement Reference	Test Method
Polyester fiber identification	See 3.3.1	AATCC TM20 (see 6.3)
Fiber blend	See 3.3.1	AATCC TM20A (see 6.3)
Denier	See 3.3.1	ASTM D1907/D1907M
Stable length	See 3.3.1	ASTM D5103
Needle-punched	See 3.3.2	Visual
Color	See 3.4	Visual
Urethane coating	See 3.5	ASTM D3776/D3776M (Opt C) and see 4.6.1 1/
Weight (after coating)	Table I	ASTM D3776/D3776M (Opt. C)
Thickness	Table I	ASTM D1777
Breaking strength.	Table I	ASTM D5034
Splitting resistance	Table I	See 4.6.2 & ASTM D5034
Fuzzing resistance	Table I	ASTM D3511/D3511M

1/ The weight of the urethane coating shall be determined during first article only.

4.6 Methods of testing and inspection.

4.6.1 Urethane coating weight test. The urethane coating weight shall be calculated by subtracting the final weight of the felt with the urethane coating from the weight of the felt determined prior the application of the urethane coating.

4.6.2 Splitting resistance test. The splitting resistance shall be determined using the apparatus as specified in ASTM D5034. Ten specimens of felt shall be cut 2-inch by 6-inch with a die from each sample unit. Five (5) shall be taken with 6-inch dimension parallel to the lengthwise direction and five (5) with the 6-inch dimension parallel to the crosswise direction. Split the conditioned specimens within the middle third of the thickness for a distance of approximately 2-inches from one (1) end only. Clamp one (1) of the 2-inch split portion specimen to the lower jaw of the testing machine and the other to the upper jaw so that movement of the jaws will separate the specimens to measure the splitting resistance. Start

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the machine, and record, as the splitting resistance in pounds, the average load in pounds necessary to pull the two (2) sections of the felt apart. Average the test results on the five (5) specimens cut lengthwise and the five (5) cut crosswise, respectively, and report separately as the splitting resistance in pounds per 2-inches of width.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The felt is intended for use in the manufacturing of individual airborne weapons cases.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2).
c. When first article is required (see 3.1).
d. Width of felt required (see 3.6).
e. Length required if other than specified (see 3.7).
f. Conformance inspection (see 4.3).
g. Inspection conditions (see 4.3.1).
h. Packaging (see 5.1).

6.3 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.4 Subject term (key word) listing.

Ammunition felt
Nonwoven

6.5 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army - GL
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Air Force - 11

Preparing Activity:
DLA -CT

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Review activities:
Army - MD

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